

Capacitors & RC Time Constants

A from-basics lesson on charge, the capacitor laws, and how voltages rise and fall in time

Day: Mon 15 Jun 2026 (catch-up) • *Phases 1–3 complete*

About this lesson

- **Sources merged (companion volumes):** *AoE* 3rd ed. §1.4–1.4.2 “Capacitors and ac circuits” (pp. 17–24, theory) • *LAoE* 2nd ed. §2N.1–2N.2 “Capacitors / Time-domain view of RCs” (pp. 55–59, discovery & intuition) • *LAoE* §2W.2 “RC step response” (pp. 109–111, worked numbers) • *Make:E* 3rd ed. Experiment 5 “Let’s Make a Battery” (pp. 37–40, hands-on) + the **RC bench lab**.
- **How the merge works:** each idea is taught *once*, from whichever book explains it best. *AoE* carries the theory; *LAoE*’s analogies and worked numbers *reinforce* it (never repeat it); *Make:E* and the bench supply the hands-on labs.
- **Length:** target ~60% of the combined in-scope source length (foundational refresher; §2/§3.6 of the Contract).
- **Year-2 modules this unlocks:** ELEC2310 Signal Processing (RC is the simplest filter/averager), ELEC2320 Control (first-order lag, the time constant τ), ELEC2322 Electronic & Computer Systems (RC timing, debouncing, logic delays).

Before you start — prerequisites are assumed, not re-taught (Skills Tracker Entry 1)

This lesson assumes the fundamentals below are already solid. Each is reminded in one line where first used and *pointed* to its full write-up — it is not re-explained here.

- **SI prefixes & unit arithmetic** (k, m, μ , n, p; carrying units) — [FT §2]
- **Rate of change / derivative** dV/dt — [FT §3]
- **Exponential function** e^x (slope = value) — [FT §4]
- **Natural log & log scale** (ln, log axis) — [FT §5]
- **First-order differential equation** (slope $\propto$ value $\Rightarrow$ exponential) — [FT §6]
- **Frequency & period** ($f = 1/T$) — [FT §7]
- **Square wave / pulse / step** — [FT §8]
- **Ohm’s law, dividers, Thévenin** ($V = IR$; collapsing a network) — [cond: AoE_1_2_2_to_1_3] / [FT §1]

1. What is a capacitor?

Curiosity hook. A resistor’s current depends on the voltage *across it right now*. A capacitor is stranger: connect a fresh one across a battery and a big gulp of current flows at the instant of connection — then dies away to nothing, even though the battery voltage never changed. Sit it in a steady (dc) circuit and it does *nothing at all*. So what is it actually doing, and why is it useless in dc but fascinating the moment anything changes?

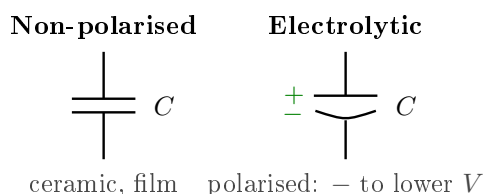


Fig. 1. Capacitor symbols (Convention Card §3). *Non-polarised*: two equal straight plates, no marks — fit either way. *Electrolytic*: flat plate = +, curved plate = –; the curved (–) plate must sit at the lower potential. (AoE Fig. 1.27)

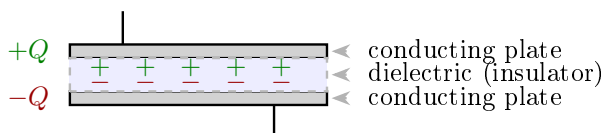


Fig. 2. Structure (LAoE Fig. 2N.1, the “ham sandwich”): two conducting plates separated by a thin insulating *dielectric*. Equal and opposite charge ($+Q$, $-Q$) sits on the two plates; the dielectric never lets charge cross.

ELI5. A capacitor is two metal plates held very close together but *not touching*, with a thin insulating layer (the *dielectric*) between them. You cannot push charge *through* it — the insulator blocks that — but you *can* pile electrons onto one plate, which shoves an equal number off the plate opposite. So one plate ends up with charge $+Q$ and the other with $-Q$. The more you pile on, the higher the voltage between the plates climbs. A capacitor is a **bucket for charge**.

Formal. A capacitor stores charge Q on its plates in direct proportion to the voltage V across them. The constant of proportionality is the **capacitance** C , measured in farads (F). Capacitance grows with plate area and shrinks with plate spacing, and is boosted by the dielectric: for an ideal parallel plate of area A and spacing d (in cm), $C = 8.85 \times 10^{-14} \varepsilon A/d$ farads, where ε is the dielectric constant (air ≈ 1 , polyester ≈ 3.1 , some ceramics ≈ 45). You almost never compute this — you buy a capacitor with the value printed on it — but it tells you *why* real capacitors are foils rolled up tightly (huge area, tiny spacing).

Key law 1 — the static (charge) relation (AoE eq. 1.13; LAoE eq. 2N.1)

$$\boxed{Q = CV} \quad \Longleftrightarrow \quad C = \frac{Q}{V}, \quad V = \frac{Q}{C}.$$

Q = stored charge (coulombs, C) V = voltage across the plates (volts, V) C = capacitance (farads, F).

One farad is enormous: in practice you meet microfarads ($\mu\text{F} = 10^{-6}$), nanofarads ($\text{nF} = 10^{-9}$) and picofarads ($\text{pF} = 10^{-12}$) [FT §2].

Types you will actually handle (for the symbol & the lab)

Ceramic (small values, pF– μF): *non-polarised* — fit either way round; use C0G/NP0 or X7R. **Film / mylar** (rolled plastic, e.g. $0.1\mu\text{F}$): *non-polarised*. **Electrolytic / tantalum** (large values, many μF , power-supply smoothing): *polarised* — the curved/marked plate is negative and **must** go to the lower voltage, or it can fail. The symbol in Fig. 1 encodes exactly this distinction.

Tiny example. A $0.1\mu\text{F}$ capacitor charged to 9V holds $Q = CV = (0.1 \times 10^{-6})(9) = 0.9\mu\text{C}$ of charge on each plate ($+0.9\mu\text{C}$ on one, $-0.9\mu\text{C}$ on the other).

Self-test 1

You double the voltage across a fixed capacitor. What happens to the stored charge? — It doubles: $Q = CV$ is a straight-line (linear) relation, so twice the voltage means twice the charge.

2. The dynamic law: current is the *rate of change* of voltage

Curiosity hook. Push a steady 1 mA into a 1 μF capacitor and its voltage does not jump — it *climbs at a steady rate* of 1000 volts per second. Hold that current for just 10 ms and the voltage has ramped up by 10 V. No resistor behaves like this. Where does that behaviour come from?

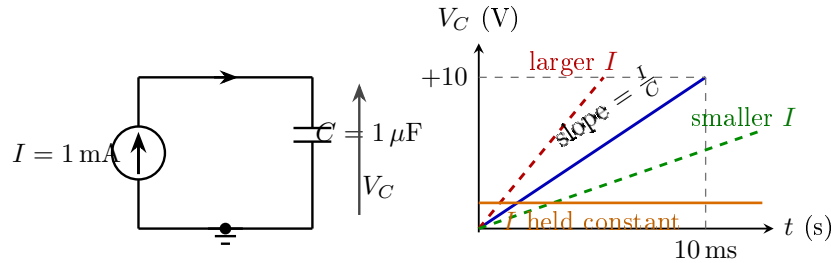


Fig. 3. A *constant* current I into a capacitor makes V_C climb in a straight-line *ramp* (AoE Fig. 1.29; LAoE Fig. 2N.3). Here 1 mA into 1 μF for 10 ms lifts V_C by 10 V. The slope is I/C : larger $I \Rightarrow$ steeper, smaller $I \Rightarrow$ shallower.

ELI5. Think of LAoE’s picture: the capacitor is a **tub of water** standing on the bench (Fig. 2N.2). The *charge* is the water in it; the *voltage* is the water level (height); the *current* is the flow rate of the hosepipe filling it; and the *capacitance* is the tub’s cross-sectional area. Fill through a thin straw (small current) and the level creeps up slowly; fill with a fire hose (big current) and it rockets up. A wide tub (big C) needs far more water to raise its level by the same amount, so it fills more slowly for a given flow. Current does not set the *level*; current sets *how fast the level changes*.

Formal. Differentiate $Q = CV$ with respect to time. Since C is constant and current is the rate of charge flow, $I = dQ/dt = C dV/dt$. So a capacitor’s current is proportional to the *time-derivative* [FT §3] of its voltage, not to the voltage itself. Run the logic backwards and it says: to change a capacitor’s voltage you must supply current, and the voltage cannot change *instantly* (that would need infinite current).

Key law 2 — the dynamic (current) relation (AoE eq. 1.15; LAoE eq. 2N.1)

$$I = C \frac{dV}{dt} \quad \Longleftrightarrow \quad \frac{dV}{dt} = \frac{I}{C}.$$

I = current into the capacitor (A) dV/dt = rate of change of capacitor voltage (V/s) C = capacitance (F).

Plain words (LAoE): “*the bigger the current, the faster the cap’s voltage changes.*”

Worked example 2a (the hook, checked). $I = 1 \text{ mA}$ into $C = 1 \mu\text{F}$:

$$\frac{dV}{dt} = \frac{I}{C} = \frac{1 \times 10^{-3} \text{ A}}{1 \times 10^{-6} \text{ F}} = 1000 \text{ V/s.}$$

Over a 10 ms pulse: $\Delta V = \frac{dV}{dt} \cdot \Delta t = 1000 \times (10 \times 10^{-3}) = 10 \text{ V. } \checkmark$

Worked example 2b (LAoE’s numbers). A 1 k Ω resistor charging a 1 μF cap from +10 V: at the first instant the full 10 V is across the resistor, so $I = V/R = 10 \text{ mA}$ and $dV/dt = I/C = 10 \times 10^{-3}/1 \times 10^{-6} = 10 \text{ V/ms}$. Later, when only 5 V remains across the resistor, $I = 5 \text{ mA}$ and the climb has *halved*. (This “starts fast, then loses its nerve” behaviour is the seed of the exponential in §5.)

Discovery / shape to recognise — the ramp (LAoE §2N.2, Fig. 2N.3–2N.4)

A **constant current** into a capacitor gives a **constant** dV/dt — a straight-line voltage ramp. This is the *signature* of “capacitor + current source.” Switch between a positive and a negative charging current and the ramp goes up, then down, then up: that is exactly how a **triangle-wave generator** works (Fig. 2N.4). True current sources are rare, though, so the *usual* case — charging through a resistor — comes in §5.

Self-test 2

*Can the voltage across a capacitor jump instantly from 0 to 5 V? — No. $I = C dV/dt$; an instant jump means $dV/dt = \infty$, which needs infinite current. Real capacitor voltages always change *smoothly*.*

3. Energy stored in a capacitor

Hook. Charge a big capacitor and short it with a screwdriver — you get a spark and a *bang*. Where was that energy hiding? Unlike a resistor, the capacitor never got hot while charging.

ELI5 → formal. As you charge a capacitor you do work pushing charge onto plates that already repel the next bit of charge. That work is not turned into heat (the capacitor is, ideally, lossless); it is stored in the electric field between the plates, and you get *all of it back* on discharge. Adding up the work ($dU = V I dt$, integrated from empty to full) gives a clean result:

Key law 3 — stored energy (AoE eq. 1.16)

$$U_C = \frac{1}{2} C V^2 \quad U_C \text{ in joules (J), } C \text{ in farads, } V \text{ in volts.}$$

A capacitor *stores* energy (in its field) and *returns* it; a resistor *dissipates* energy (as heat) and cannot.

Tiny example. A $10\mu\text{F}$ cap charged to 5 V stores $U = \frac{1}{2} C V^2 = \frac{1}{2} (10 \times 10^{-6}) (5)^2 = 1.25 \times 10^{-4} \text{ J} = 125 \mu\text{J}$ — modest, but a 1 F supercap at 5 V stores 12.5 J, enough to bite.

4. Combining capacitors

Visual-first idea. Capacitors combine the *opposite* way to resistors. Putting two in *parallel* is like widening one big tub — more area, so more charge at the same voltage: capacitances **add**. Putting two in *series* stacks the plates so the same charge sits on a larger effective spacing — the combination stores *less*.

Combining rules (AoE eqs. 1.17–1.19)

Parallel (capacitances add, like series resistors):

$$C_{\text{total}} = C_1 + C_2 + C_3 + \dots$$

Series (combine like parallel resistors):

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots \quad \text{or, for two,} \quad C_{\text{total}} = \frac{C_1 C_2}{C_1 + C_2}.$$

Why series works (AoE Exercise 1.15). The middle node between two series capacitors has no

external wire, so whatever charge $+Q$ leaves the bottom plate of C_1 must arrive as $-Q$ on the top plate of C_2 : *the two capacitors carry equal charge*. The total voltage is $V = Q/C_1 + Q/C_2$, and dividing Q through gives the reciprocal-sum rule.

Self-test 3

Two $1\,\mu\text{F}$ capacitors in series — what is the total? — $\frac{C_1 C_2}{C_1 + C_2} = \frac{1 \times 1}{1 + 1} = 0.5\,\mu\text{F}$. In parallel they would give $2\,\mu\text{F}$.

5. RC circuits in time: discharging

Curiosity hook. Charge a capacitor, then connect it across a resistor and walk away. The voltage does not fall in a straight line and it does not fall off a cliff — it sags in a particular curved way, fast at first and ever slower, the *same shape* whatever the values. That shape is everywhere in engineering. Why this shape, and not some other?

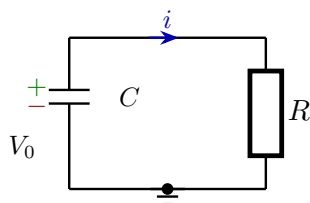


Fig. 4. The simplest RC circuit (AoE Fig. 1.31): a capacitor charged to V_0 shares a loop with a resistor R . With the source removed, C drives a current i through R and discharges.

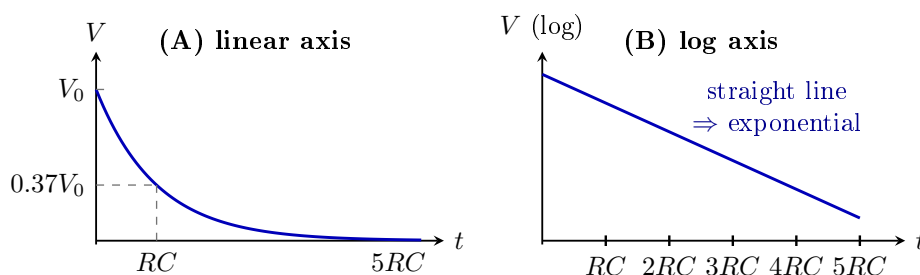


Fig. 5. Discharge curve (AoE Fig. 1.32; LAoE Fig. 2N.6). **(A)** On a linear axis $V = V_0 e^{-t/RC}$ falls to **37 %** of V_0 in one RC , and is essentially zero by $5RC$. **(B)** The *same* decay drawn on a logarithmic V axis is a *straight line* ([FT §5]) — a quick test for a true exponential.

ELI5 — the intuition (LAoE). Two everyday pictures. (i) **Draining bathtub:** a full tub drains fast because deep water pushes hard through the plughole; as it empties, the shallow water pushes feebly, so it drains ever more slowly — it never *quite* reaches empty. (ii) **Zeno’s paradox:** the runner covers half the remaining distance, then half of that, then half again — always approaching, never arriving. The capacitor does the same: the current that drains it (through R) is proportional to the voltage *still left*, so as the voltage falls the draining slows in step.

Formal — why it must be an exponential. Two facts collide. Ohm’s law on the resistor: the draining current is $I = V/R$. The capacitor law: that current sets the slope, $C\,dV/dt = -I$ (minus because V is falling). Put them together:

$$C \frac{dV}{dt} = -\frac{V}{R} \quad \Longrightarrow \quad \frac{dV}{dt} = -\frac{1}{RC} V.$$

This is a **first-order differential equation** [FT §6]: “the rate of change is proportional to the value itself.” The only function whose slope is a constant times itself is the exponential [FT §4] — so the solution is forced:

Discharge (AoE eq. 1.20)

$$V(t) = V_0 e^{-t/RC}$$

V_0 = starting voltage; t = time (s); RC = the time constant (see §7). At $t = RC$, $e^{-1} = 0.37$, so V has fallen to **37%** of V_0 .

6. RC circuits in time: charging

Hook. Now do the opposite: at $t = 0$ flick a switch that connects a battery V_f through a resistor to an empty capacitor. The voltage climbs — but as a mirror image of the discharge, fast at first then easing as it nears V_f , which it approaches but (in theory) never quite reaches.

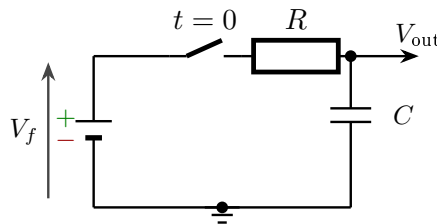


Fig. 6. RC charging circuit (AoE Fig. 1.33). Closing the switch at $t = 0$ connects the battery V_f through R to C . The output, taken across C , climbs from 0 toward V_f .

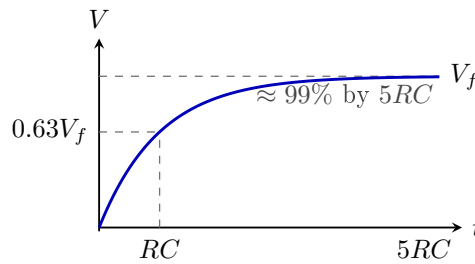


Fig. 7. Charging curve (AoE Fig. 1.34; LAoE Fig. 2N.6): $V = V_f(1 - e^{-t/RC})$ rises toward the dashed final value V_f , reaching **63 %** of V_f in one RC and $\approx 99\%$ by $5RC$.

Formal. Same two laws, but now the current is driven by the voltage *still missing* across the resistor, $(V_f - V)/R$:

$$C \frac{dV}{dt} = \frac{V_f - V}{R} \quad \Rightarrow \quad V(t) = V_f(1 - e^{-t/RC}).$$

Charge (AoE eq. 1.22; LAoE §2N.2)

$$V(t) = V_f(1 - e^{-t/RC})$$

At $t = RC$, V has covered **63%** of the gap to V_f (because $1 - e^{-1} = 0.63$).

The two numbers worth memorising, and the data table (LAoE §2N.2, Fig. 2N.6)

LAoE: “*in one RC the cap goes 63% of the way to its destination; in five RCs, 99% of the way.*”
In full:

Elapsed time	0	1τ	2τ	3τ	4τ	5τ
Charging — % of V_f reached	0%	63%	86%	95%	98%	99%
Discharging — % of V_0 remaining	100%	37%	14%	5%	2%	1%

(Charging and discharging are mirror images: “% reached” + “% remaining” = 100 at every instant.)

The 5τ rule of thumb (AoE §1.4.2B)

A capacitor charges or decays to within **1% of its final value in five time constants**. For most practical purposes, “after 5τ it has arrived.”

7. The time constant $\tau = RC$

ELI5 → **formal**. The whole speed of an RC circuit is set by one number: the product $\tau = RC$. Bigger R (a thinner straw) or bigger C (a wider tub) both make everything slower. Remarkably, R in ohms times C in farads comes out directly in *seconds*.

Time constant

$$\tau = RC \quad (\text{seconds, for } R \text{ in } \Omega \text{ and } C \text{ in F})$$

Unit shortcuts [FT §2]: $\text{M}\Omega \times \mu\text{F} \rightarrow \text{seconds}$; $\text{k}\Omega \times \mu\text{F} \rightarrow \text{milliseconds}$.

Example: $1\text{ k}\Omega \times 1\text{ }\mu\text{F} = 1\text{ ms}$; and at 1 V the initial current is $V/R = 1\text{ mA}$.

 τ as a “ballpark” — and a trick for the calculator-less (LAoE §2N.2)

You quote τ even when you never plug it into the formula: it tells you *how quick or slow* a circuit is. See fuzz at 1 MHz on the scope? You instantly know not to blame an RC whose τ is in the millisecond range.

No calculator? The remaining “error” (gap to the destination) shrinks to about **40%** ($\approx 1/e$) of itself *each* time constant: $1 \rightarrow 0.4 \rightarrow 0.16 \rightarrow 0.06 \rightarrow \dots$ — which reproduces the 63/86/95/98% table above.

8. Useful timing relationships

Rearranging the charge/discharge equations [FT §5] gives the formulae you actually reach for at the bench.

Solve for the time to reach a chosen voltage (AoE eq. 1.22)

Charging toward V_f , the time to reach a voltage V is

$$t = RC \ln\left(\frac{V_f}{V_f - V}\right).$$

- **The 50% point:** a charging (or discharging) RC reaches *half-way* at $t = RC \ln 2 \approx \mathbf{0.7 RC}$. (Used for logic time-delays, §10.)

- **Rise time (10% → 90%):** $t_r = RC \ln 9 \approx 2.2 RC$ (AoE Exercise 1.16).

Worked example 8a. How long for a $1\text{ k}\Omega/1\text{ }\mu\text{F}$ circuit ($\tau = 1\text{ ms}$) charging from 0 toward 5 V to reach 2.5 V (half-way)?

$$t = RC \ln \frac{5}{5 - 2.5} = 1\text{ ms} \times \ln 2 = 0.69\text{ ms}.$$

Exactly the $0.7 RC$ rule. ✓

9. Driving an RC with a square wave (toward the step response)

Hook. Feed an RC not one step but a *square wave* [FT §8] — up, down, up, down. Each edge launches a fresh charge or discharge. What the output looks like depends entirely on how the wave’s timing compares with τ .

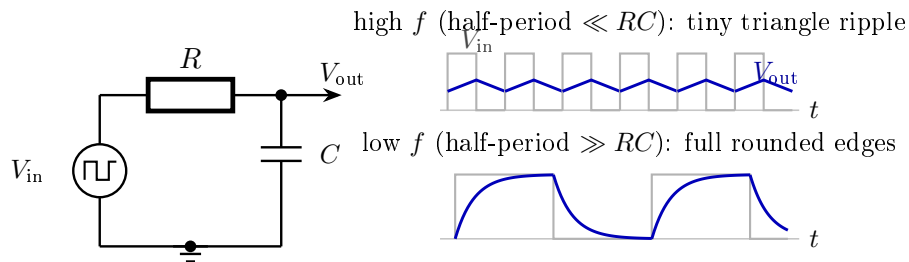


Fig. 8. A square wave through R into C (AoE Fig. 1.35). When the half-period is much *shorter* than RC the cap barely moves, so V_{out} is only a small triangular ripple. When it is much *longer*, the cap fully charges and discharges each half-cycle, tracing rounded exponential edges.

The rule of thumb that ties it together: compare the half-period of the drive to τ . If there is plenty of time ($\gg \tau$) the capacitor reaches the rails; if there is little time ($\ll \tau$) it hardly charges and the output stays near its average. This single comparison — τ *versus* how long the input holds still — is the key to the worked step-response next.

10. Worked example: the RC step response (LAoE §2W.2)

The problem (LAoE Fig. 2W.10). A 10 V, $10\text{ }\mu\text{s}$ pulse is fed through a $0.01\text{ }\mu\text{F}$ coupling capacitor into a “next stage.” Sketch the output for different next stages. *The lesson is in how RC compares with the $10\text{ }\mu\text{s}$ pulse width.*

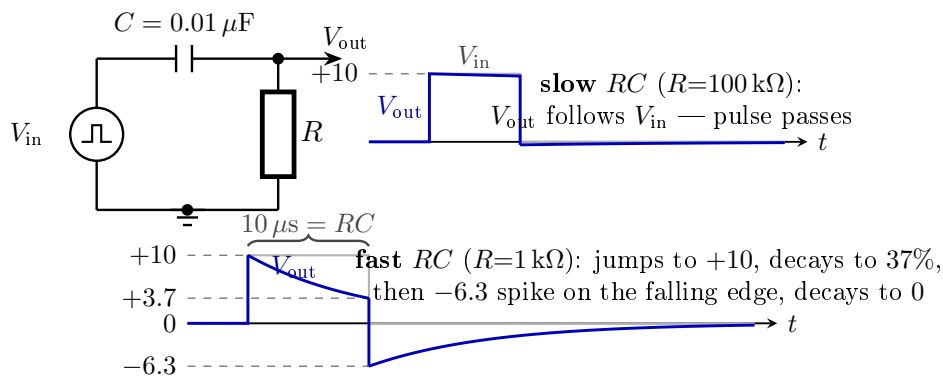


Fig. 9. Pulse (step) response of the series- C / shunt- R high-pass (LAoE Figs. 2W.13–2W.14), pulse = +10 V for $10\text{ }\mu\text{s}$. **Slow** RC ($1\text{ ms} \gg 10\text{ }\mu\text{s}$): the cap barely charges, so the pulse passes almost unchanged. **Fast** RC ($\approx 10\text{ }\mu\text{s}$): the edges are differentiated — a +10 V spike decaying to $\approx 37\%$, then a $\approx -6.3\text{ V}$ spike decaying back to 0.

The step-response take-away

- $RC \gg$ pulse width $\Rightarrow$ capacitor barely charges $\Rightarrow$ **output follows input** (the pulse “passes”).
- $RC \approx$ pulse width $\Rightarrow$ **exponential decay is visible**; the output falls to 37% in one τ (the same $e^{-t/RC}$ as §5).

This is the same physics as the square-wave of §9, now read off a single pulse.

Honest scope note (LAoE footnote)

The pure-capacitor “next stages” above are idealised: a capacitor with *no* dc path to ground would slowly charge up on leakage current. A real circuit always provides a resistive path to ground to define the dc level. (The deeper ac-coupling/blocking story is *parked* — see the box at the end.)

Self-test 4 — combines two ideas

A $1\text{ k}\Omega + 1\text{ }\mu\text{F}$ RC ($\tau = 1\text{ ms}$) is driven by a 50 Hz square wave. Does the output look like clean exponential edges or a small triangle ripple? — A 50 Hz square wave holds each level for 10 ms, which is $10\tau \gg \tau$, so the cap fully charges and discharges each half-cycle: **rounded exponential edges** reaching the rails (the “slow RC / pulse passes” case).

11. Taming complicated RCs, and a real timer

Thévenin to the rescue (AoE Fig. 1.36). A capacitor fed from a *divider* (R_1 over R_2) looks complicated, but it is not: collapse the divider to its Thévenin equivalent [`cond: AoE_1_2_2_to_1_3`] — one source V_{th} in series with one resistance $R_{th} = R_1 \parallel R_2$ — and you are back to a plain single- R RC, with $\tau = R_{th}C$.

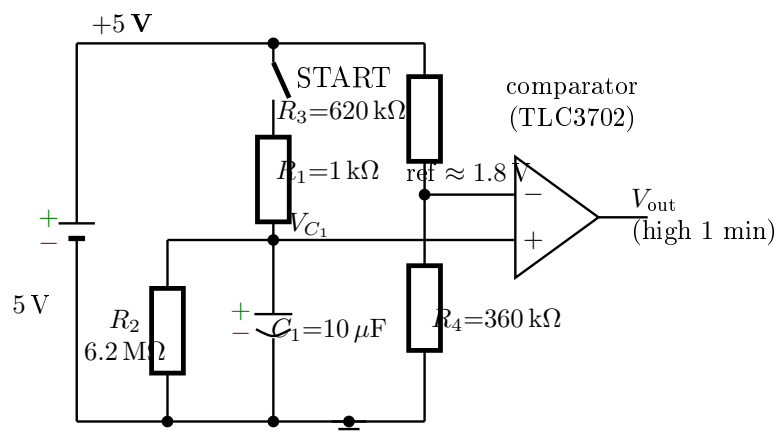


Fig. 10. “One Minute of Power” timer (AoE Fig. 1.38, simplified). START charges C_1 *fast* through R_1 ($\tau = 10\text{ ms}$) to +5 V; on release it discharges *slowly* through R_2 ($\tau = R_2 C_1 \approx 62\text{ s}$). The comparator holds its output *high* while V_{C_1} stays above the divider reference ($\approx 1.8\text{ V} = 1/e$ of 5 V); since reaching $1/e$ takes exactly one time constant, the output stays high for about one minute.

Why 37%? (the elegant bit). The reference is set to $1/e \approx 37\%$ of the supply *on purpose*: a discharge reaches $1/e$ of its start after exactly *one* time constant (§5). So choosing $\tau = R_2 C_1 \approx 1\text{ min}$ makes the timer run for one minute, by construction.

Inrush, via Key law 2. Omitting R_1 would still “work,” but connecting an empty cap straight across

5 V means trying to make 5 V of dV in ≈ 0 s of dt — a huge current spike ($I = C dV/dt$). The series R_1 limits the peak charging current to a modest ≈ 5 mA. This is Key law 2 used as a design tool.

Lab A — Build a battery from a lemon (Make:E Experiment 5)

Why this lab is here

Every RC circuit needs a dc source to charge the capacitor. Before trusting the bench supply, *make* a source from scratch and feel where “voltage” comes from — a chemical push of charge. A student who has lit an LED from four lemons never again thinks of a battery as a mystery box.

You will need: 2 lemons (or a squeeze-bottle of pure lemon juice); 4 copper-plated coins (UK/US pennies — *new and bright*; oxidised dull-brown ones work poorly); 4 galvanised (zinc-plated) steel brackets/mending plates, $\geq 1''$ (silver-grey finish); test leads (1 red, 1 black, 3 others); a multimeter; a generic red LED.

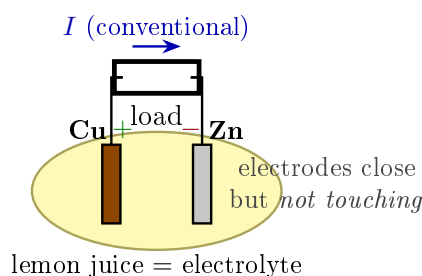


Fig. 11. A single electrochemical cell (Make:E Figs. 5-1/5-4): a copper electrode (the + terminal) and a zinc electrode (the – terminal) pushed into lemon-juice electrolyte, close but *not touching*. Conventional current leaves the copper + terminal, drives the load, and returns to the zinc –.

Procedure — Part one: one cell.

1. Cut a lemon in half. Push a bright penny into it. As close as possible *but not touching*, push in a galvanised plate.
2. Set the multimeter to the 2 V dc range. Hold the **red** probe on the penny, the **black** probe on the plate.
3. *What you should see:* between **0.8 V and 1 V** from this single cell.

Procedure — Part two: light the LED. One cell cannot reach an LED’s forward voltage, so stack cells in *series* — more lemons.

1. Make 4 cells (4 lemons, or a divided parts-box with juice in each compartment).
2. Link them with alligator leads so each lead joins a *plate to a penny* (a zinc of one cell to the copper of the next). **Never** connect penny–penny or plate–plate.
3. Connect the LED across the two free ends (long leg to the copper/+ end). *What you should see:* the LED glows. (Use 4 cells rather than 3: the LED loads the battery and pulls its voltage down, and the cells’ *high internal resistance* keeps the current tiny — safe, but it needs the headroom.)

The physics — “The Nature of Electricity” (Make:E)

A battery is **electrochemical**: chemistry pushes charge. In each cell a reaction between zinc and lemon juice pulls positive *zinc ions* out of the zinc electrode, leaving electrons stranded on the metal. Those electrons (which repel one another) escape by running through the external wire and the load to the copper electrode, losing a little energy in the load on the way. At the copper, positive *hydrogen ions* in the juice grab the arriving electrons and bubble off as hydrogen gas.

Primary vs secondary: this is a *primary* cell — it makes current the instant you connect it, and is dead once the electrode metal is consumed (not easily rechargeable). A *secondary* (rechargeable) cell uses electrodes/electrolyte whose reactions can be reversed by an external charging current.

Conventional current — why arrows point $+\rightarrow-$ (Make:E)

Electrons actually drift from $-$ to $+$. But Franklin (1747) had guessed current flowed $+$ to $-$, and J. J. Thomson only found the electron in 1897 — by then the convention was a century old, and the maths works identically either way (negative charge moving one way $\equiv$ positive charge moving the other). So we keep **conventional current**: **from $+$ to $-$** . The arrows inside diode and transistor symbols point this way.

Cleanup. The hardware is reusable though discoloured. *Do not eat the lemons* — metal has been deposited in them.

Lab B — Measure a time constant on the bench (RC bench lab)

The point

Everything in §5–§8 is a prediction. Here you *watch* a capacitor charge, mark the 63% point with your own eyes, and check that the measured τ matches RC . Predict first, then measure — that is the whole game.

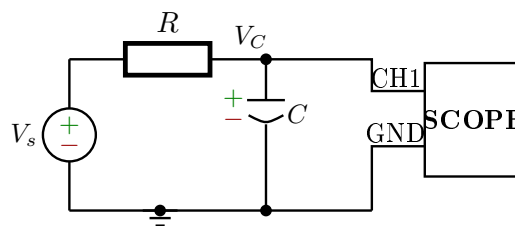


Fig. 12. RC bench setup: the DC bench supply drives R into C ; the node V_C is the output. The scope’s CH1 probe sits on V_C and its ground clip returns to the same signal ground, so it reads the capacitor voltage as it charges.

Kit: Siglent SPD3303X PSU (or a square-wave source), Rigol DHO924S scope, Extech EX530A DMM, a breadboard. Components: $R = 10\text{ k}\Omega$ and $C = 100\text{ }\mu\text{F}$ (slow); $R = 1\text{ k}\Omega$ and $C = 1\text{ }\mu\text{F}$ (fast). Use metal-film 1% resistors; use an *electrolytic* for $100\text{ }\mu\text{F}$ (mind polarity!), a film/ceramic for $1\text{ }\mu\text{F}$.

Predict first.

$$\tau_{\text{slow}} = RC = (10 \times 10^3)(100 \times 10^{-6}) = 1\text{ s}; \quad \tau_{\text{fast}} = (1 \times 10^3)(1 \times 10^{-6}) = 1\text{ ms}.$$

Procedure — the slow circuit (watch it by eye/DMM).

1. Build $10\text{ k}\Omega$ in series with $100\text{ }\mu\text{F}$ to ground. Set the PSU to 5.00 V , current limit 50 mA , output *off*.
2. Put the DMM across the capacitor. *Predict*: on switch-on, V_C should reach 63% of $5\text{ V} = 3.15\text{ V}$ after one $\tau = 1\text{ s}$, and be within 1% of 5 V after $5\tau = 5\text{ s}$.
3. Switch the output on and start a timer. *What you should see*: $\approx 3.15\text{ V}$ at 1 s , $\approx 4.3\text{ V}$ at 2 s , essentially 5 V by 5 s — the table of §6, live.

Procedure — the fast circuit (on the scope).

1. Build $1\text{ k}\Omega$ in series with $1\text{ }\mu\text{F}$. Drive the input with a low-frequency square wave (e.g. 100 Hz , $0\text{--}5\text{ V}$) so each half-period (5 ms) is $\gg \tau = 1\text{ ms}$ (the “slow RC / pulse passes” case of §9).
2. Scope on V_C . *What you should see*: rounded exponential charge/discharge edges reaching the rails.
3. **Measure τ** : read the time for V_C to climb from 0 to 63% of 5 V ($= 3.15\text{ V}$) using the scope cursors. *Predict 1 ms .*

Measured vs calculated — close the loop

Tabulate, for each circuit: calculated $\tau = RC$, measured τ (time to the 63% point), and the % difference. Real electrolytics carry wide tolerance ($\pm 20\%$ is common) and resistors $\pm 1\%$, so a measured τ within $\sim 10\text{--}20\%$ of RC is a *pass* — and the gap is mostly the capacitor’s tolerance, not an error in the physics. (This mirrors LAoE **Lab 2L**, where the same curve is first met on the scope.)

Schematic symbol reference (this lesson)

Capacitor symbols — per the Schematic Conventions Reference

- **Non-polarised** (ceramic, film): *two equal, straight, parallel plates*; no polarity marks; connect either way.
- **Electrolytic / polarised**: *one straight plate (positive) and one curved (or shorter) plate (negative)*, with explicit $+$ and $-$ marks. The curved/marked plate is the negative terminal and must sit at the lower potential.

(Drawn for real in Fig. 1, Phase 2. No new *circuit* symbols beyond these are introduced by this lesson.)

What is NOT in this lesson (parked — so nothing is silently dropped)

Parked topics and where they live

- **Differentiators & integrators** (RC that outputs dV/dt or $\int V dt$) — AoE §1.4.3–1.4.4, LAoE §2N.2.1 (Figs. 2N.7–2N.9). *Parked* $\rightarrow$ *tomorrow’s lesson*.
- **The frequency-domain view** (reactance $X_C = 1/\omega C$, impedance, phasors, RC high/low-pass *filters*, the 3 dB point, 90° phase) — AoE §1.7, LAoE §2N.3–2N.6 and §2W.1. *Parked* $\rightarrow$ *the AoE §1.7 day*. (The ac-coupling/“blocking” use of a capacitor, LAoE §2N.4, rides along with that.)

This lesson is deliberately the *time-domain* story only: charge, the two laws, and how voltages rise and fall in time.

Schedule cross-reference

Source (in scope)	Where it landed in this lesson
AoE §1.4 intro	§1 hook; “useless in dc, interesting when changing”
AoE §1.4.1 Capacitors	§1 ($Q = CV$, structure, types, symbol), §2 ($I = C dV/dt$), §3 (energy), §4 (combining)
AoE §1.4.2 RC: V and I vs time	§5 (discharge), §6 (charge), §7 (τ), §8 (timing), §9 (square wave), §11 (Thévenin, timer)
LAoE §2N.1 Capacitors	§1 (ham sandwich, static/dynamic), §2 (tub analogy)
LAoE §2N.2 Time-domain view	§2 (ramp), §5 (Zeno, bathtub), §6 (% table), §7 (ballpark, calc-less trick); Lab 2L $\rightarrow$ Lab B
LAoE §2W.2 RC step response	§10 (worked pulse cases)
Make:E Experiment 5	Lab A (lemon battery, electrochemistry, conventional current)
RC bench	Lab B (10k+100 μ F, 1k+1 μ F, 63% point, measured vs calc.)

Self-test 5 — “what would happen if...”

You rebuild Lab B’s fast circuit but accidentally fit 10 μ F instead of 1 μ F. With the same 100 Hz square wave, what changes on the scope? — Now $\tau = RC = (1\text{ k})(10\text{ }\mu\text{F}) = 10\text{ ms}$, while each half-period is only 5 ms ($\approx \frac{1}{2}\tau$). The cap no longer reaches the rails: instead of full rounded edges you see a smaller, more triangular ripple riding near the mid-voltage (the “fast square wave / barely charges” case of §9).

Self-test 6

Why does a discharge plotted on a logarithmic voltage axis come out as a straight line? — Taking $\ln$ of $V = V_0 e^{-t/RC}$ gives $\ln V = \ln V_0 - t/RC$, a straight line in t with slope $-1/RC$ [FT §5]. The log axis “undoes” the exponential, which is why decay curves and frequency responses are usually drawn that way.

v1.3 (19 Jun 2026): Fig. 3 “slope = I/C ” label rotated to sit along the blue ramp it describes (was floating upper-left). No content, value or coverage changes.

v1.2 (overseer cosmetic pass, 19 Jun 2026): inline fractions $\text{dfrac} \rightarrow \text{frac}$ in running text (display blocks keep dfrac); Fig. 3 “slope = I/C ” label compacted and moved clear of the ramps, “I held constant” nudged (figure data unchanged); title-block subtitle “Phase-1 draft — diagrams pending” $\rightarrow$ “Phases 1–3 complete”; preamble comment $v2.10 \rightarrow v2.12$. No content, value or coverage changes.

v1.1 (final-review fix, 19 Jun 2026): the five tfrac fractions (the §3 energy formula and its tiny example, the §9 self-test, and the coverage-ledger energy row) changed to dfrac to match the house standard; no content, value, figure or wording changes.

v1.0 (Phases 1–3 complete): lesson text, worked examples, data tables and labs (Pass-1 coverage ledger below); all 12 figures drawn in circuitikz/TikZ and verified by Pass-2 (Convention-Card check, per-coordinate junction-dot audit, wire-continuity trace, 200 dpi inspection); 19 new terms folded into the Keywords & Definitions MASTER. Ready for sign-off.

Appendix — Pass 1 Coverage Ledger (Content-Conservation Accounting, Contract §2)

Every in-scope source heading, key equation, worked example and figure is mapped to a location in this lesson, or marked parked. All 12 figures were drawn and audited in Phase 2.

Source item	Status	Location / note
<i>AoE §1.4–1.4.2 (theory)</i>		
§1.4 intro (useless in dc)	✓	§1 hook
§1.4.1 $Q = CV$ (eq. 1.13)	✓	§1, Key law 1
§1.4.1 structure / parallel-plate (eq. 1.14)	✓	§1 (Fig. 2, dielectric note)
§1.4.1 dielectric constants, types	✓	§1 “Types” note
§1.4.1 symbol (Fig. 1.27)	✓	§1 Fig. 1 + symbol-reference box
§1.4.1 $I = C dV/dt$ (eq. 1.15)	✓	§2, Key law 2
§1.4.1 ramp example (Fig. 1.29)	✓	§2 Fig. 3, worked ex. 2a
§1.4.1 energy $U = \frac{1}{2}CV^2$ (eq. 1.16)	✓	§3, Key law 3
§1.4.1 photo of types (Fig. 1.30)	omit	component-variety photo; distilled to §1 “Types” (not padding-critical)
§1.4.1 parallel/series (eqs. 1.17–1.19)	✓	§4 (incl. Ex. 1.15 equal-charge reasoning)
§1.4.2 time vs frequency framing	✓	§5 intro; frequency parked (end box)
§1.4.2 discharge ODE & soln (eq. 1.20)	✓	§5, Key law (discharge)
§1.4.2 discharge curves, lin+log (Fig. 1.31/1.32)	✓	§5 Figs. 4–5 (37% @ τ)
§1.4.2 $\tau = RC$, 1 k·1 μ F = 1 ms	✓	§7
§1.4.2 charging (eq. 1.21–1.22, Fig. 1.33/1.34)	✓	§6 (63% @ τ)
§1.4.2 5 RC rule	✓	§6 note
§1.4.2 $t = RC \ln(V_f/(V_f - V))$	✓	§8
§1.4.2 rise time 2.2 RC (Ex. 1.16)	✓	§8
§1.4.2 averaging weight $e^{-\Delta t/RC}$	note	§9 (mentioned as “stays near average”; full averaging $\rightarrow$ freq. day)
§1.4.2 square-wave drive (Fig. 1.35)	✓	§9 Fig. 8
§1.4.2 Thévenin reduction (Fig. 1.36, Ex. 1.17)	✓	§11
§1.4.2 time-delay, 50%@0.7 RC (Fig. 1.37)	✓	§8 (0.7 RC), §11
§1.4.2 “One Minute of Power” (Figs. 1.38–1.40)	✓	§11 Fig. 10 + inrush note; output-driver zoo (LED/piezo/relay/SSR/triac) distilled to one line
§1.4.3 Differentiators / §1.4.4 Integrators	parked	end box $\rightarrow$ tomorrow
<i>LAoE §2N.1–2N.2 (discovery / labs)</i>		
§2N.1.1 Why? (delay, filter)	✓	§1 hook; filter parked
§2N.1.2 structure (ham sandwich, Fig. 2N.1)	✓	§1 Fig. 2
static $Q = CV$ / dynamic $I = C dV/dt$	✓	§1–§2 framing
hydraulic / tub analogy (Fig. 2N.2)	✓	§2 ELI5
constant- I ramp (Fig. 2N.3)	✓	§2 Fig. 3 + ramp box
triangle-wave generation (Fig. 2N.4)	✓	§2 ramp box
exponential charging, Zeno, bathtub (Fig. 2N.5)	✓	§5 ELI5 + worked ex. 2b
63% / 99%, % table (Fig. 2N.6)	✓	§6 table
$e^{-1} = 0.37$; M $\Omega\mu$ F $\rightarrow$ s, k $\Omega\mu$ F $\rightarrow$ ms	✓	§6, §7

Source item	Status	Location / note
“time constant” as ballpark	✓	§7
calculator-less “error $\rightarrow$ 40% each	✓	§7
τ ” trick		
Lab 2L (scope observation)	✓	cross-ref in Lab B
§2N.2.1 integrators & differentiators (Figs. 2N.7–9)	parked	end box $\rightarrow$ tomorrow
§2N.3–2N.6 frequency domain, phasors, filters	parked	end box $\rightarrow$ AoE §1.7 day
<i>LAoE §2W.2 (worked step response)</i>		
§2W.2.1 problem: pulse via coupling cap (Fig. 2W.10)	✓	§10 setup
case $R=\infty, C=0$ (passes) (Fig. 2W.11)	✓	§10 take-away (“passes”)
case C–C divider $0.9V_{in}$ (Fig. 2W.12)	note	§10 (mentioned as attenuated-but-passes; C–C divider detail is light)
case $R=100k$ ($RC \gg$ pulse, passes) (Fig. 2W.13)	✓	§10 Fig. 9 (slow RC)
case $R=1k$ (decays to 37%) (Fig. 2W.14)	✓	§10 Fig. 9 (fast RC)
case $R=100k, C=0.01\mu F$ (half amp, Fig. 2W.15)	note	§10 (covered by “passes at reduced amplitude”)
no-dc-path footnote	✓	§10 honest scope note
§2W.1 RC filters (frequency)	parked	end box $\rightarrow$ AoE §1.7 day
<i>Make:E Experiment 5</i>		
build: lemon/penny/zinc, 0.8–1 V/cell	✓	Lab A part one
series stacking $\rightarrow$ light LED (Figs. 5-1/5-2)	✓	Lab A part two
nature of electricity, ions, H_2 (Figs. 5-3/5-4)	✓	Lab A physics note
primary vs secondary battery	✓	Lab A physics note
conventional current (Franklin/Thomson)	✓	Lab A conventional-current box
cleanup	✓	Lab A cleanup
<i>RC bench</i>		
10 k+100 $\mu F \rightarrow \tau = 1$ s	✓	Lab B slow circuit
1 k+1 $\mu F \rightarrow \tau = 1$ ms	✓	Lab B fast circuit
63% point; measured vs calculated	✓	Lab B procedures + close-the-loop box
τ		

Worked-example count. Source worked instances in scope: AoE ramp (1.29), energy (1.14), series-cap derivation (1.15), Thévenin RC (1.17), rise-time (1.16), the timer (1.38); LAoE 10 V/ms numbers, the % table, the §2W.2 pulse cases. This lesson carries worked examples 2a, 2b, 8a, the §10 pulse cases, six self-tests, plus both labs — $\geq 80\%$ of source worked content retained (Golden Rule).